[Elbauenpark Magdeburg]

Can SAT Solvers Keep Up With the Linux Kernel's Feature Model?

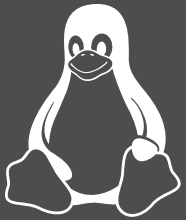
ICSE 2026 — April 12–18 — Rio de Janeiro, Brazil

Elias Kuiter[1], Urs-Benedict Braun[1], Thomas Thüm[2], Sebastian Krieter[2], Gunter Saake[1]

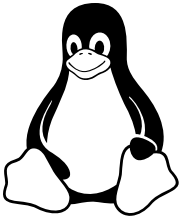
University of Magdeburg[1], Braunschweig[2], Germany

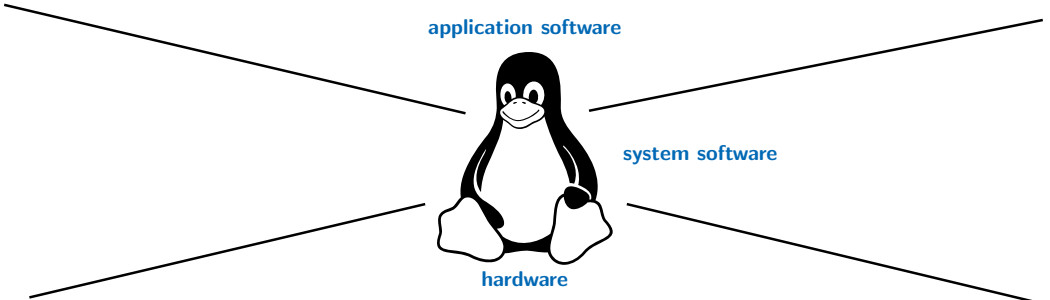
OTTO VON GUERICKE UNIVERSITÄT MAGDEBURG

Technische Universität Braunschweig



Motivation and Background




application software
system software
hardware

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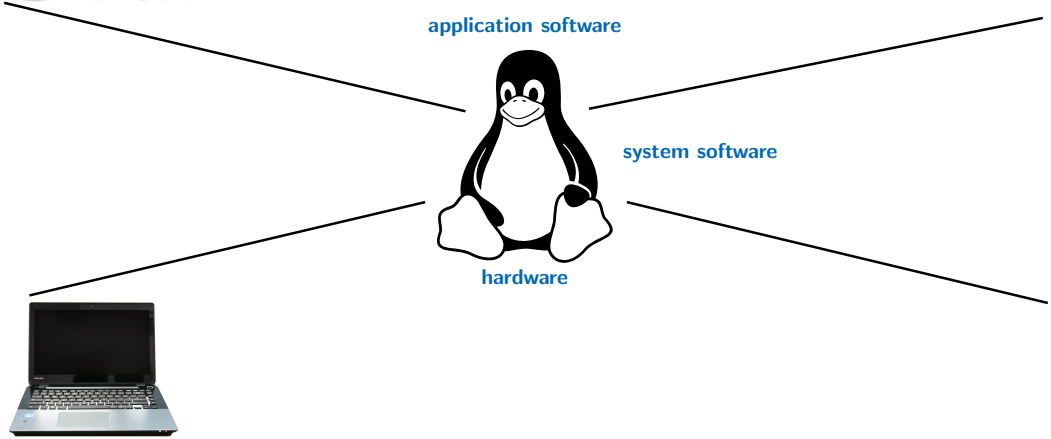

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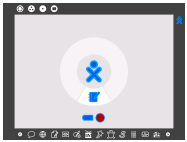

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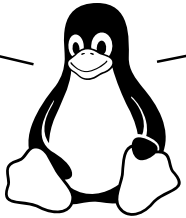

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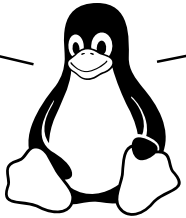

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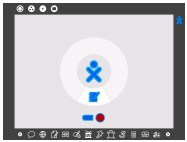

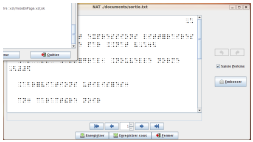

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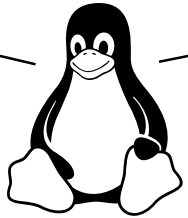

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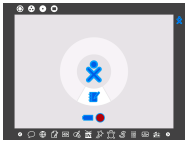

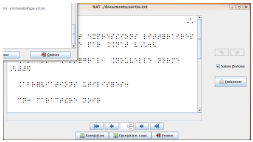

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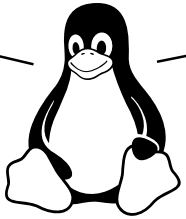

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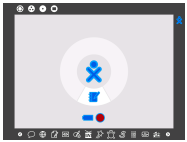

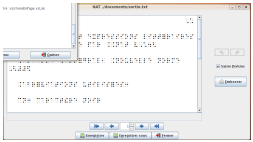

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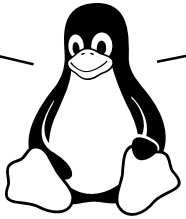

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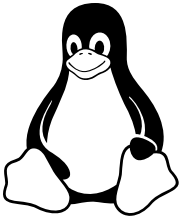

KConfig Feature Model ...

```
config TOSHIBA
  tristate "Toshiba Laptop"
  depends on X86_32
  help
    Toshiba portables with a
    genuine Toshiba BIOS
```

```
config OLPC
  bool "One Laptop Per Child"
  depends on !X86_PAE
  help
    Unique features of the OLPC
    XO hardware
```

```
config SUPERH
  def_bool y
  select ARCH_32BIT_OFF
  help
    Sega Dreamcast gaming
    console
```

```
config X86_32_IRIS
  tristate "Eurobraille/Iris"
  depends on X86_32
  help
    Iris machines from
    EuroBraille
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config X86_GOLDFISH
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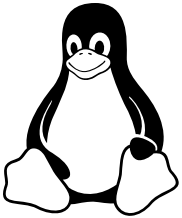
Feature-to-Code Mapping



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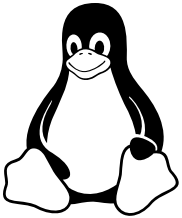
KBuild Makefiles



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C Preprocessor

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#ifdef CONFIG_TOSHIBA
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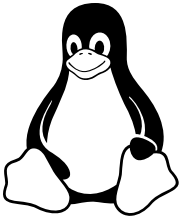

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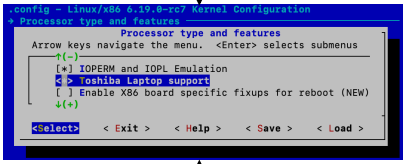
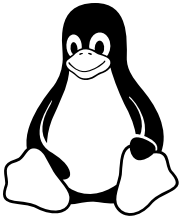
make menuconfig





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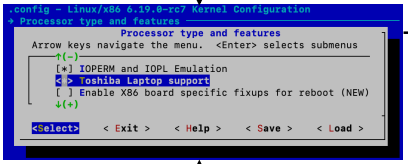

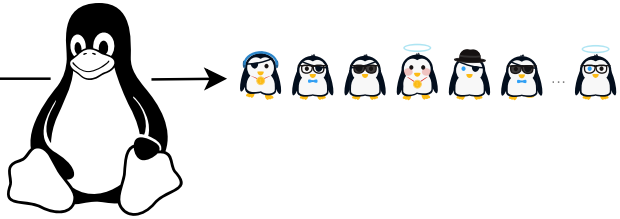

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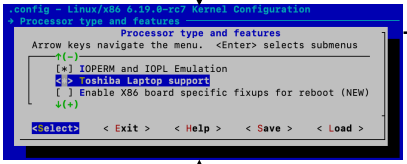

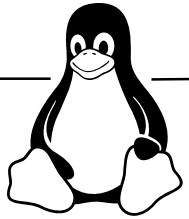

Linux is a

software product line

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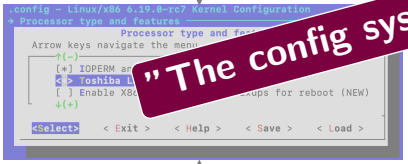

"The config system is a nightmare"

Linux is a
software product line

KConfig Feature Model

```
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  help
    Toshiba portables with a
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    _smm(&regs);
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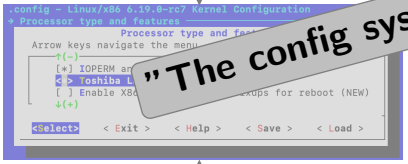

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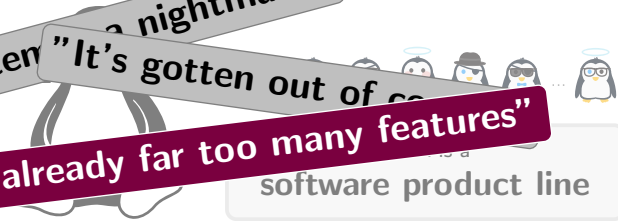

"The config system is a nightmare"

"It's gotten out of control"

"There are simply already far too many features"

software product line

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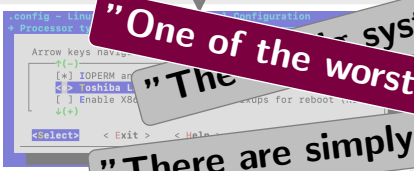

"One of the worst parts of the whole project" [Linus Torvalds]

"There are simply alre… …so many features"

"It's gotten out of co…"

"…a nightmare"

"…d system…"

"The…"

…product line

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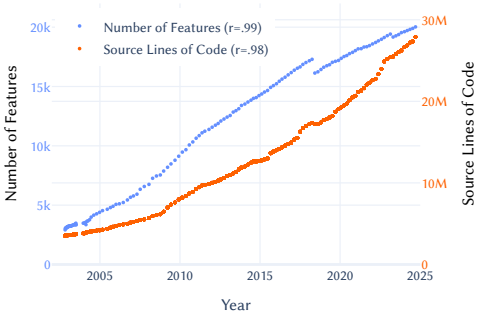
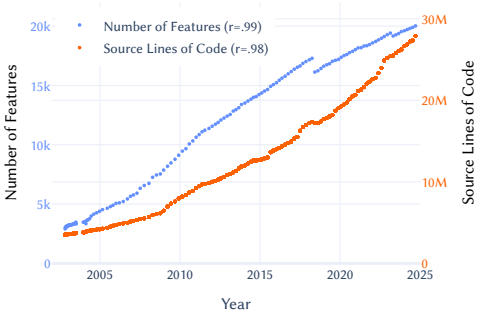
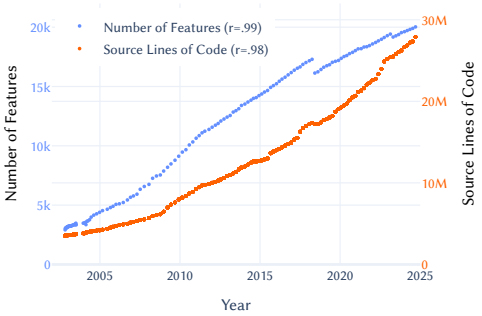

Number of Features (r=.99)
Source Lines of Code (r=.98)
Number of Features
Source Lines of Code
Year
20k
15k
10k
5k
0
30M
20M
10M
0
2005
2010
2015
2020
2025



The Linux kernel grows **more complex** ...



interactive product configuration anomaly detection and explanation feature-model evolution and slicing modularization implicit constraints code parsing dead code analysis code simplification type checking model checking formal verification static code analysis dataflow analysis testing t-wise sampling uniform random sampling optimization of non-functional properties refactoring economical estimations ...

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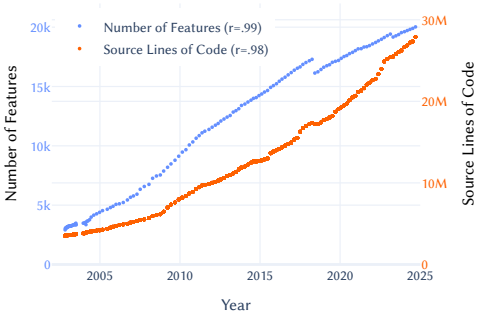

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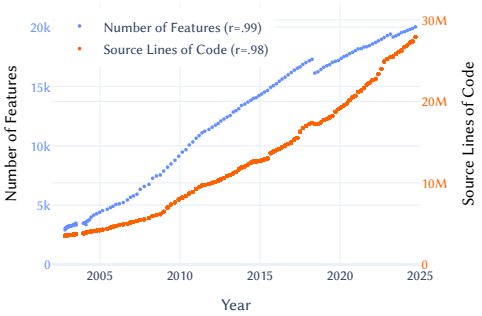
... and so do **dozens of analyses**.

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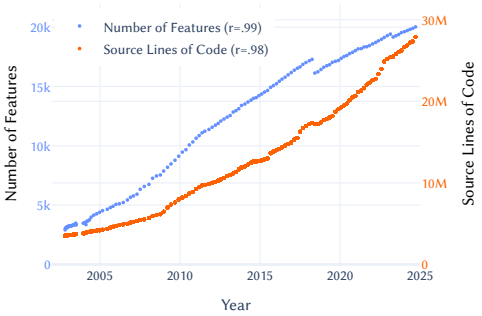


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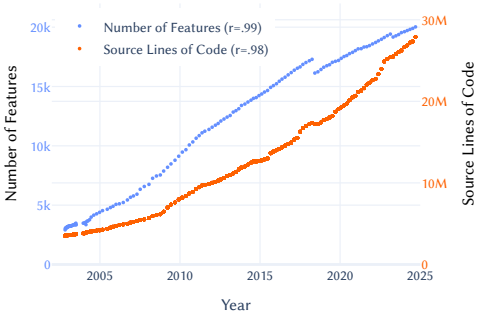
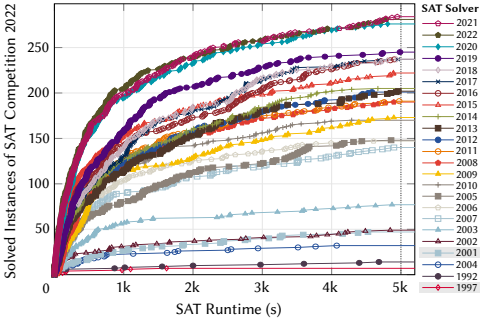
... relying on **feature models** ...

... relying on **SAT solvers**.





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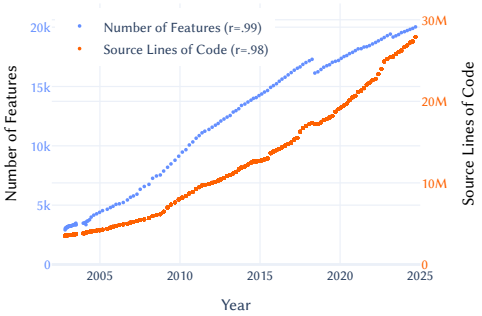
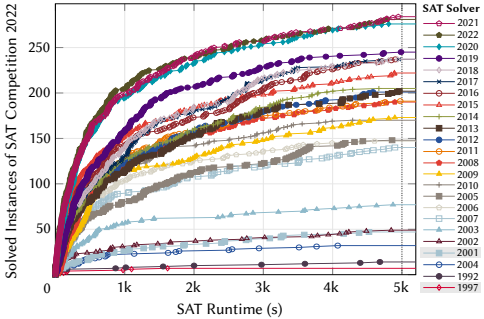
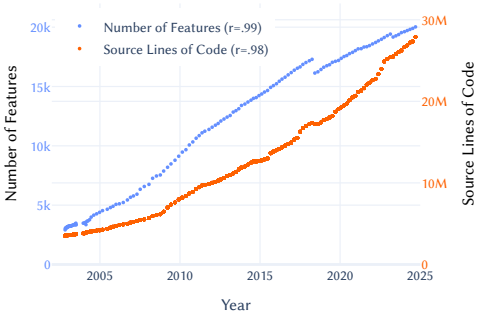




The Linux kernel grows **more complex** ...

... yet SAT solvers **get faster** every year.

Linux Kernel vs. SAT Solvers



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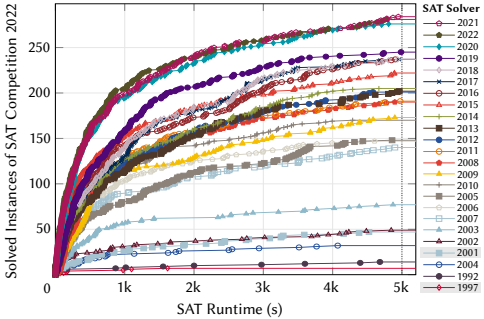

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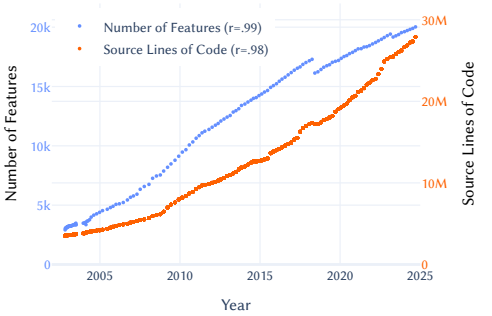

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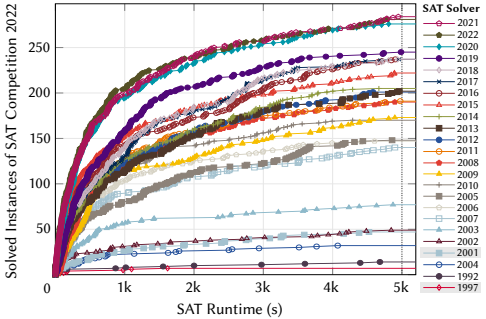

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Can SAT solvers **keep up** with the Linux kernel?

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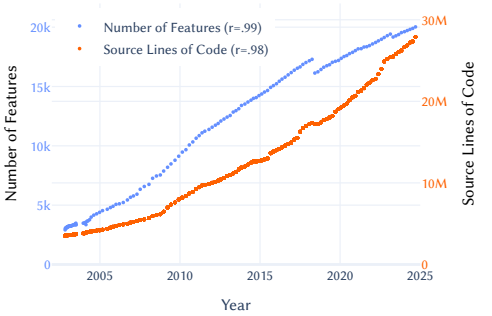

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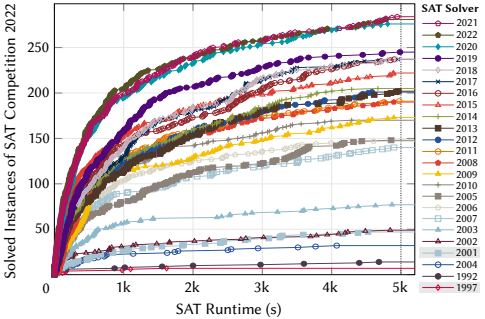

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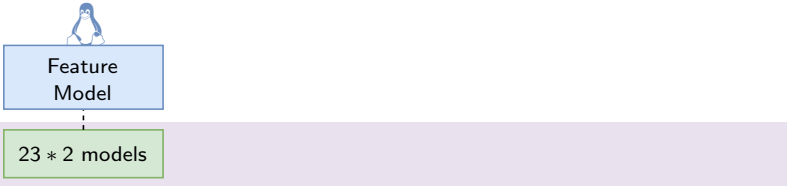
Can SAT solvers **keep up** with the Linux kernel? – Spoiler: **They cannot!**


How to evaluate this?

Research Questions and Evaluation Pipeline

Research Questions and Evaluation Pipeline

RQ$_2$ How does SAT runtime **evolve** on the history of Linux?



Years 2002–24

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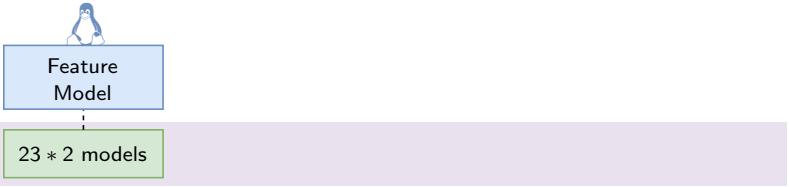

Years 2002–24
`x86` and `arm`

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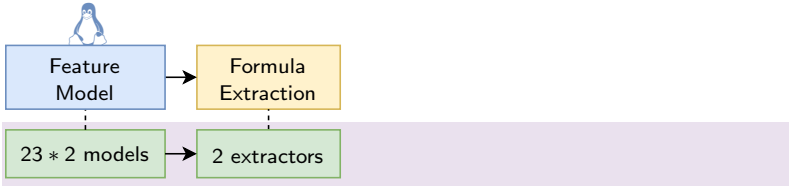

Years 2002–24	KConfigReader
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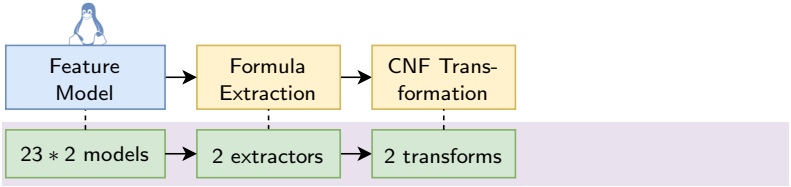

Years 2002–24	KConfigReader	KConfigReader
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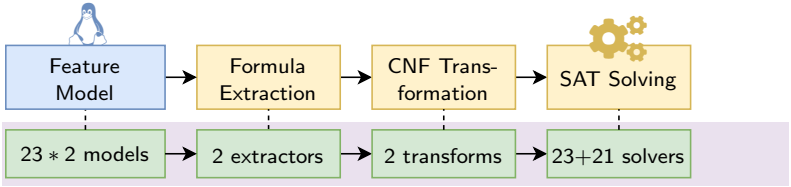

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`x86` and `arm`	KClause	Z3	SAT Museum

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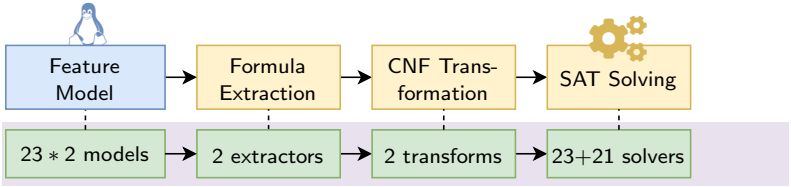

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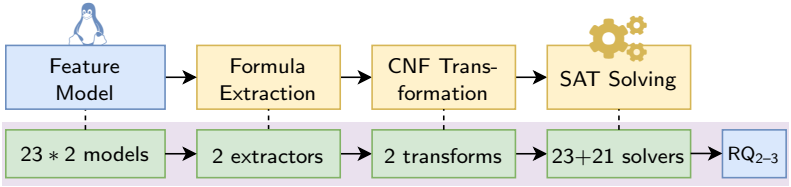

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Measured Metric

Runtime of a SAT Query

Research Questions and Evaluation Pipeline

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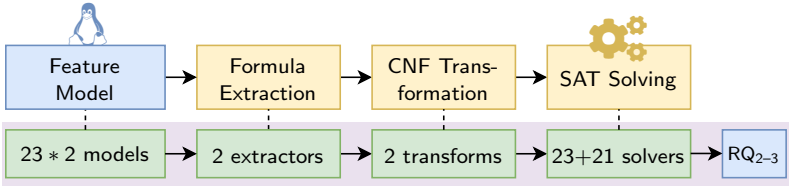

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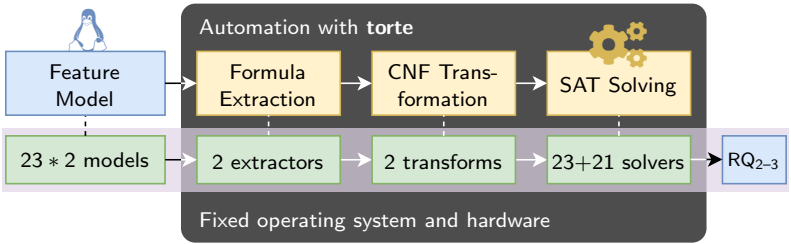

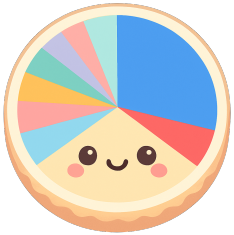

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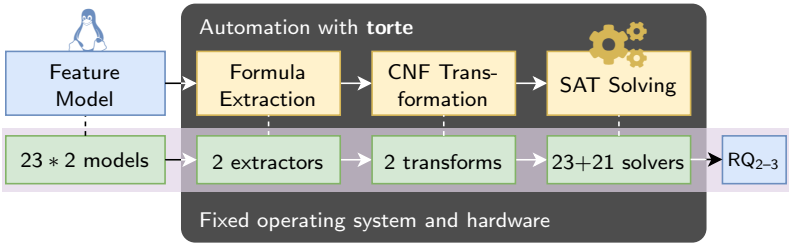
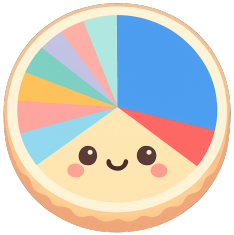




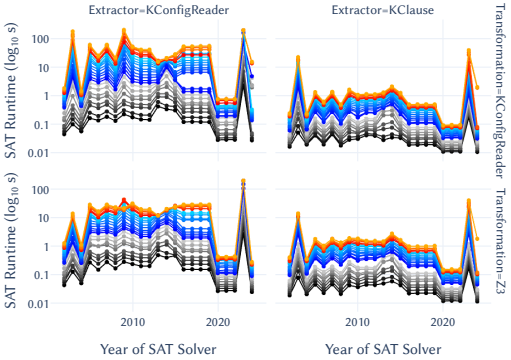
Years 2002–24	KConfigReader	KConfigReader	SAT Competition
`x86` and `arm`	KClause ∗5	Z3 ∗5	SAT Museum ∗3

Measured Metric

Runtime of a SAT Query

RQ_2: How Does ⚙ SAT Runtime Evolve on 🐧 Linux?

RQ_2: How Does SAT Runtime Evolve on Linux? (logarithmic y-axis!)



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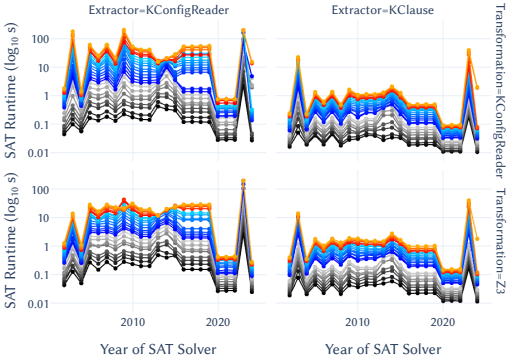

Every Line Is a 🐧 Feature Model . . .

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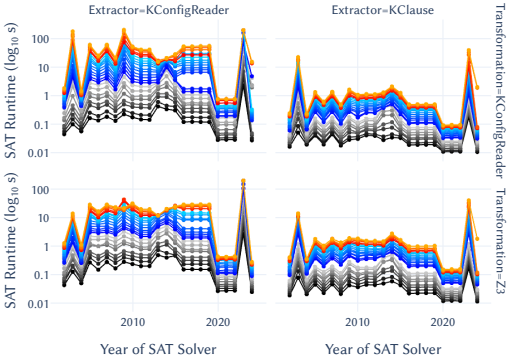

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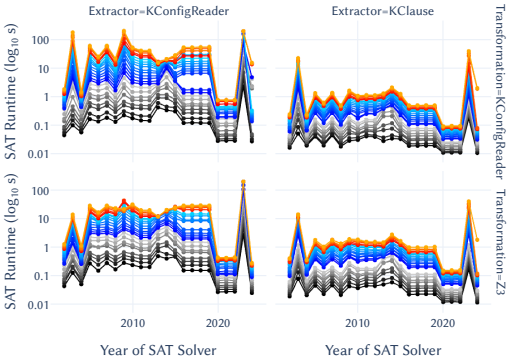

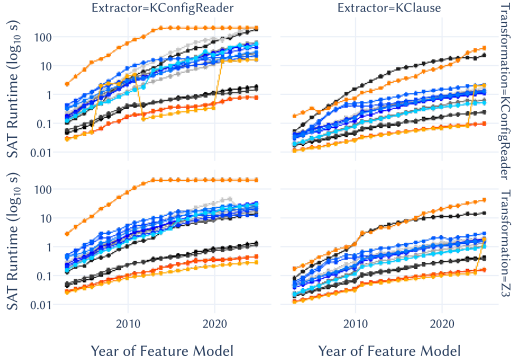

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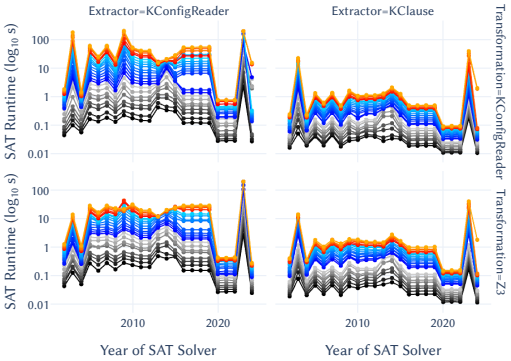

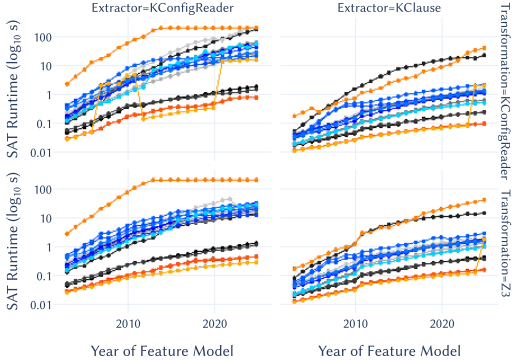

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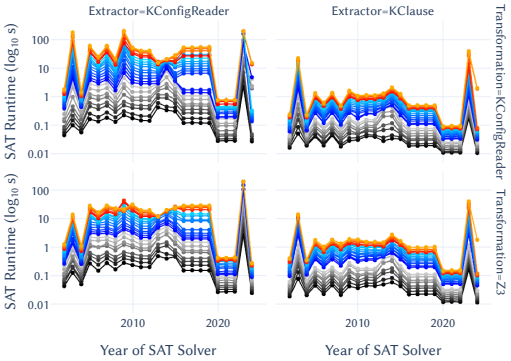

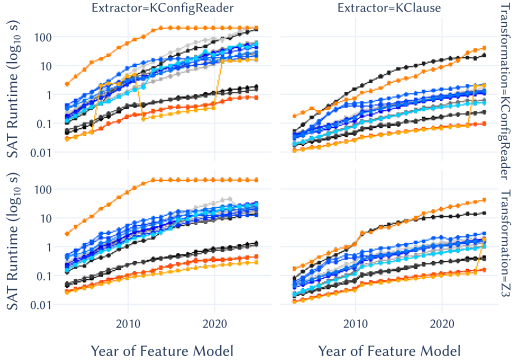

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Every Line Is a SAT Solver ...

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RQ_3: How Do Optimal ⚙ Solvers Evolve on 🐧 Linux?

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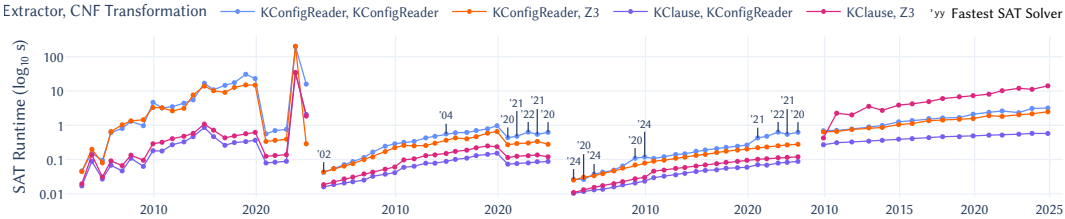

(a) Current Strategy ($RQ_{3.1}$) (b) Optimal Strategy ($RQ_{3.2}$) (c) Historic Strategy ($RQ_{3.3}$) (d) Pragmatic Strategy (RQ_4)

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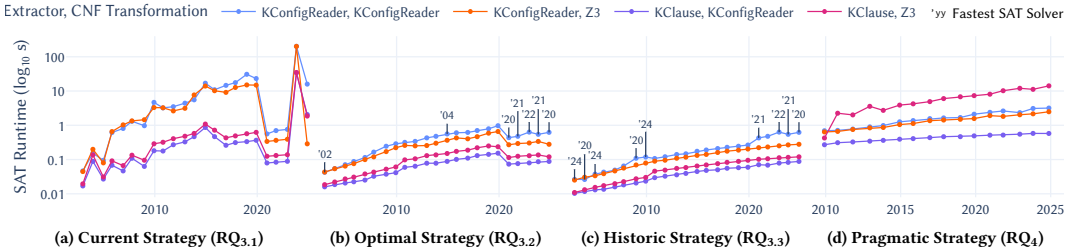

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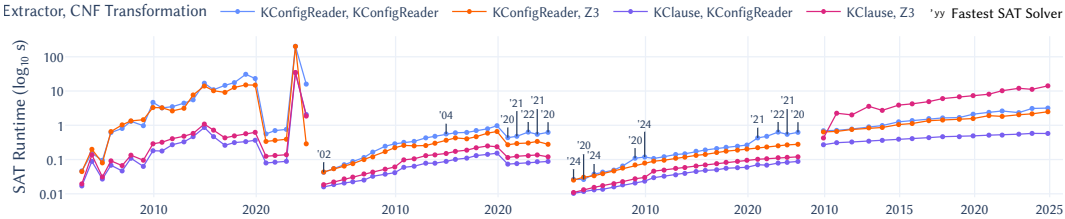

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* despite disruptive innovations like `Kissat`

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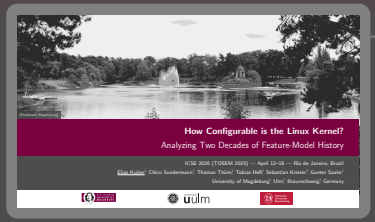
Thank you for listening!

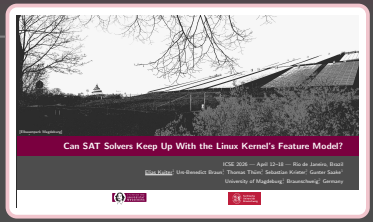
Conclusion

Journal-First Paper



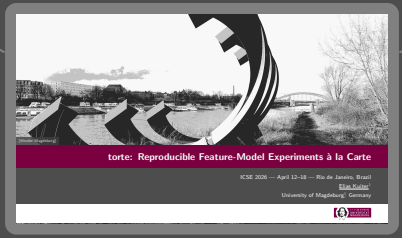
How Configurable Is the Linux Kernel?

Research Paper



Can SAT Solvers Keep Up With the Linux Kernel's Feature Model?

Demo Paper



torte: Reproducible Feature-Model Experiments à la Carte

Experiment Variables

Variable	Type	Values	RQ
Feature Model			
Version	Independent	*(cf. Table 2a)*	$RQ_{2–4}$
Architecture	Independent	`x86`, `arm`	$RQ_{1.1}$
Formula Extraction	Independent	KConfigReader, KClause	$RQ_{1.2,\ 2–4}$
CNF Transformation	Independent	KConfigReader, Z3	$RQ_{1.3,\ 2–4}$
SAT Solving			
Solver	Independent	*(cf. Table 2b)*	$RQ_{2–4}$
Compiler (only for ①)	Confounding?	Competition: `gcc/g++ ??`	$RQ_{1.4}$
	Control	Museum: `gcc/g++ 9.4.0`	
Analysis	Control	SAT query (20min timeout)	—
Deep Variability			
Operating System	Control	Ubuntu `22.04` (in Docker)	—
Hardware	Control	Intel Xeon E5-2630 2.4 GHz	—
Iteration	Independent	$5 * 5 * 3$ iterations	$RQ_{1.5,\ 2}$
SAT Runtime	Dependent	*(cf. Section 5)*	$RQ_{1–4}$

Feature Models and SAT Solvers

Year	Version	Year	Version
’02	2.5.53	’14	3.18
’03	2.6.0	’15	4.3
’04	2.6.10	’16	4.9
’05	2.6.14	’17	4.14
’06	2.6.19	’18	4.20
’07	2.6.23	’19	5.4
’08	2.6.28	’20	5.10
’09	2.6.32	’21	5.15
’10	2.6.36	’22	6.1
’11	3.1	’23	6.6
’12	3.7	’24	6.12
’13	3.12	*’25*	*N/A*

In experiment ①, we analyze all years. In experiment ②, we only analyze years ’09–’24. In each analyzed year, we consider two architectures (`x86`, `arm`). We only consider the mainline kernel of Linux.

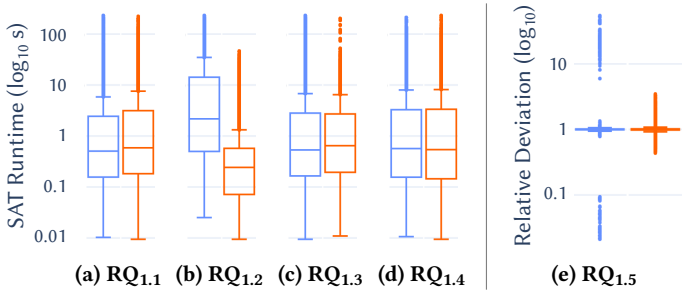
(a) Feature Models

	Year	Solver
Experiment ①	’02, ’04	`zChaff`†, `Limmat`‡
	’03	`Forklift`,† `BerkMin`‡
	’05	`SatELiteGTI`
	’06, ’08	`MiniSat`
	’07	`RSat`
	’09	`PrecoSAT`
	’10	`CryptoMiniSat`
	’11–12	`Glucose`
	’13–14	`Lingeling`
	’15	`abcdSAT`
	’16–19	`MapleSAT`
	’20–22, ’24	`Kissat` (’24†)
	’23	`SBVA-CaDiCaL`†
Exp. ②	’09–11	`Sat4j (2.1.0)`
	’11–14	`Sat4j (2.3.1)`
	≥ 2014	`Sat4j (2.3.5)`

†SAT Competition only ‡SAT Museum only

(b) SAT Solvers

RQ_1: Influence Factors



(a) $RQ_{1.1}$ (b) $RQ_{1.2}$ (c) $RQ_{1.3}$ (d) $RQ_{1.4}$ (e) $RQ_{1.5}$

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- **Sonic Adventure**, Sonic Team [https://en.wikipedia.org/wiki/File:Sonic_Adventure_Dreamcast.png]
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